

Poly(*N*-isopropylacrylamide) Self-Fabricated Coating for Marine Anti-Biofouling

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Titanium (Ti) or Ti alloy pipelines have been increasingly used in seawater cooling systems of marine facilities and offshore equipments due to their high corrosion resistance, specific strength, and fatigue resistance. However, their good biocompatibility renders them high susceptibility to marine biofouling, which can block the pipeline system and reduce its cooling efficiency. Self-fabricated coating is presented based on poly(*N*-isopropylacrylamide) (PNIPAM) to combat marine biofouling in titanium seawater pipeline systems. When PNIPAM solution in seawater is applied onto the Ti surface by flushing, PNIPAM chains would be adsorbed on the inner wall of the pipe and self-fabricate into a coating due to the coordination interactions between nitrogen in PNIPAM and Ti as well as the hydrophobic interactions from hydrocarbon moieties of PNIPAM. Self-fabrication is confirmed by quartz crystal microbalance with dissipation, X-ray photoelectron spectroscopy, and atomic force microscopy. The coating exhibits good anti-biofouling with $\approx 99\%$ antibacterial rate, while it does not reduce the corrosion resistance of titanium. The coating can serve more than 10 days at a flow velocity of 0.3 m s^{-1} and continue to work once it is regenerated by flushing the PNIPAM solution. This study provides a facile approach to protect the titanium pipeline from marine biofouling.

Ti is regarded as one of the most promising materials for next-generation marine facilities and offshore equipments. Yet, because of its excellent biocompatibility,^[8] Ti surface is prone to fouled by marine organisms such as barnacles, mussels and algae, leading to serious negative impacts on the marine equipment. In particular, biofouling in seawater pipeline systems usually blocks the pipes, valves, and heat exchangers, reducing their cooling efficiency and even posing safety risk.^[9,10]

In the past years, great efforts, including using antifouling coatings,^[11–14] surface chemical grafting,^[15–17] micro-arc oxidation,^[17–20] and antifouling agents^[21–23] have been made to mitigate the marine fouling growth on Ti surface. Antifouling coating is the most convenient and effective approach to inhibit fouling on ship hulls.^[24–27] However, such coatings do not apply to internal seawater systems such as seawater cooling pipelines, because of their limited space, complex structures, and irregular parts. Moreover, the service life of Ti-alloy equipment largely exceeds that

of the existing antifouling coatings,^[28] recoating or repairing during pipeline operation is difficult. Surface chemical grafting and micro-arc oxidation can produce stable coatings through covalent or metallic bonding,^[16,20] but its long-term antifouling performance is still questionable. Moreover, such surface treatments on a scale as large as seawater cooling pipeline is a challenge. Antifouling agents can be introduced into seawater pipelines via electrolysis generating Cu ion or NaClO or by directly adding agents such as quaternary ammonium salts or isothiazolinone-based compounds,^[21,22] and they show good antifouling efficiency in some cases.^[9] However, their release would cause pollution of the marine environment.^[29] Also, such antifouling agents must be added once a day, which is time-consuming and costly.

In this study, we present an antifouling coating fabricated by poly(*N*-isopropylacrylamide) (PNIPAM) chains on the Ti pipe surface in situ. PNIPAM with a low critical solution temperature (LCST) is well known for its thermo-sensitivity.^[30–34] On the other hand, PNIPAM chain can be adsorbed on various metal surfaces due to the coordination interactions of its nitrogen atoms with metal and hydrophobic interactions from its hydrocarbon moieties,^[35–41] where part of the chain can be free in the seawater forming a hydrated layer at a temperature below its LCST,

1. Introduction

Owing to the high corrosion resistance, specific strength, and fatigue resistance,^[1,2] titanium (Ti) and Ti alloys have found applications in seawater cooling pipelines, heat exchangers, pressure-resistant hulls, propellers, and other marine equipments,^[3–7] so

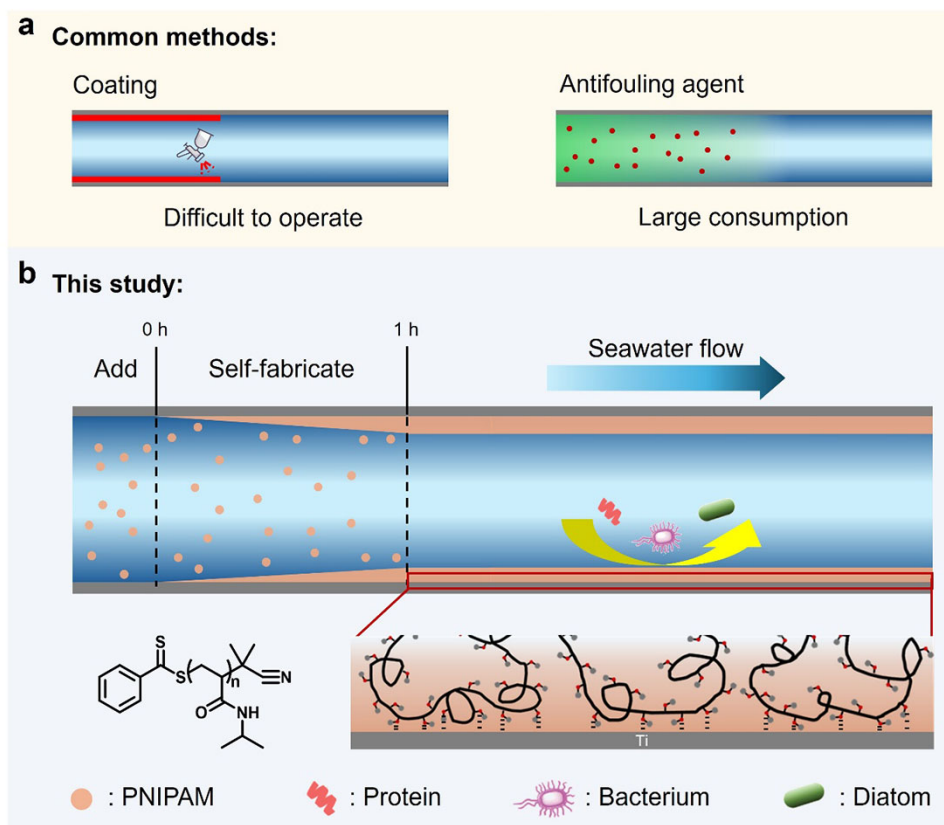
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Scheme 1. Schematic of the common antifouling methods (a) and antifouling strategy of the self-fabricated PNIPAM coating for the Ti pipeline (b).

which resists the biofouling adhesion.^[42–45] By flushing PNIPAM solution in seawater, PNIPAM chains self-fabricate into a coating on the inner wall of the titanium seawater pipe. The coating can be regenerated by the same procedure once it is washed out. We have investigated the fabrication of PNIPAM chains and the antifouling as well as anti-corrosion performances of the coating. Our aim is to develop a high-performance antifouling material for seawater pipeline systems.

2. Results and Discussion

2.1. Self-Fabrication of PNIPAM Chains

PNIPAM sample was synthesized via reversible addition-fragmentation chain transfer (RAFT) polymerization (Scheme 1). Proton nuclear magnetic resonance (¹H NMR) spectra confirmed the successful synthesis (Figure S1, Supporting Information). Table 1 shows the PNIPAM samples with different molecular weights. For convenience, PNIPAM samples are designated as LPN-*x*, where *x* is the polymer molecular weight.

The adsorption of LPN-*x* on the Ti surface was examined by using a quartz crystal microbalance with dissipation (QCM-D), where Ti-coated sensor was used. Figure 1a shows that frequency shift (Δf) and dissipation shift (ΔD) as a function of time. After the introduction of PNIPAM solution, either $-\Delta f$ or ΔD increased with time until an equilibrium was reached. After rins-

ing, for each sample, $-\Delta f > 1000$ Hz and $\Delta D > 9.0 \times 10^{-5}$. It is known that $-\Delta f$ increases with the mass on the sensor surface, whereas ΔD increases with the thickness and viscoelasticity of the polymer layer.^[46–48] Clearly, PNIPAM chains were adsorbed on the Ti surface. On the other hand, $-\Delta f$ gradually increased with the molecular weight of PNIPAM, in that the sites for coordination bonding of N atoms with Ti and those for hydrophobic interactions increase with the molecular weight.^[38,39]

Simulation was performed to examine the adsorption of PNIPAM on the Ti surface (Figure 1b). The average enthalpy change for the adsorption of a single molecule is -173.8 kcal mol⁻¹, further indicating that the adsorption is feasible.^[49]

Considering that PNIPAM coating was a viscoelastic layer composed of PNIPAM chains and coupled water,^[50] we estimated the adsorption in terms of the Voigt model. For LPN-5, LPN-11,

Table 1. Characterization of PNIPAM samples.

Polymer	Feed molar ratio [CPBD:PNIPAM]	M_n [10^3 g mol ⁻¹] ^{a)}	M_w [10^3 g mol ⁻¹] ^{a)}	PDI ^{a)}
LPN-5	1:44	5.2	6.2	1.2
LPN-11	1:88	11.3	14.7	1.3
LPN-20	1:176	20.0	22.8	1.1
LPN-40	1:352	40.1	49.0	1.2

^{a)} Determined by gel permeation chromatography.

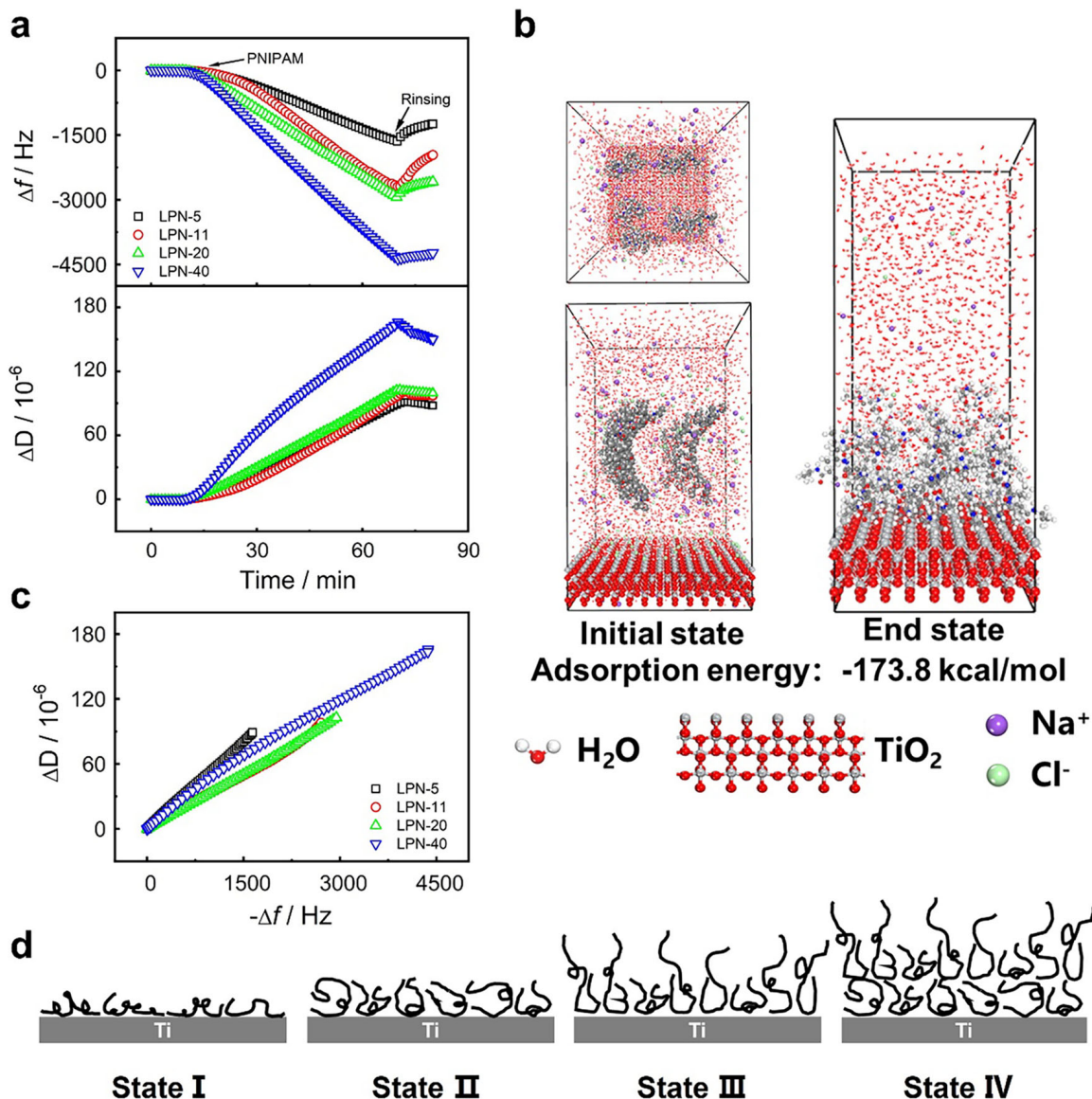


Figure 1. a) Time dependence of the frequency shift (Δf) or dissipation shift (ΔD) for the adsorption of PNIPAM on the Ti surface at 25 °C. b) Self-fabricated simulation experiment of PNIPAM on the Ti surface at NaCl solution. c) The ΔD versus $-\Delta f$ relation. d) Schematic illustration of self-fabrication of PNIPAM chains into a coating.

LPN-20, or LPN-40, it was 8.5, 13.8, 19.6, or 33.6 $\mu\text{g cm}^{-2}$, and the thickness was 71, 115, 165, or 285 nm (Figure S2, Supporting Information).

Figure 1c shows that ΔD linearly increased with $-\Delta f$, indicating that the PNIPAM coating was uniform or the PNIPAM chains were uniformly packed. Note that the slope of the line almost did not change, indicating that PNIPAM chains did not form a polymer brush.^[47] Instead, they self-fabricated into a coating with some segments attached on Ti surface due to N-Ti coordination interactions and hydrophobic interactions, and some segments or tails free in the seawater.

The adsorption of PNIPAM chains is illustrated in Figure 1d. In the initial stage, a PNIPAM chain almost lies on the Ti surface due to the coordination and hydrophobic interactions, since

each unit can act as an interacting site. As the adsorption increases (Stages II and III), some segments are still attached to Ti surface while the rest have to leave the surface and extend to the seawater, so that more PNIPAM chains are attached. Further increasing the adsorption leads the attached PNIPAM chains to fabricate with some free PNIPAM chains into coating (Stage IV), where the segments of PNIPAM chains at the top are free in the seawater, forming a hydration layer. Accordingly, such partially attached PNIPAM chains can effectively exert the antifouling performance.

The PNIPAM coating was characterized by X-ray photoelectron spectroscopy (XPS), water contact angle (WCA), and atomic force microscopy (AFM) (Figure 2). In the XPS spectrum of bare Ti (Figure 2a), peaks corresponding to O (530 eV) and Ti (458 eV)

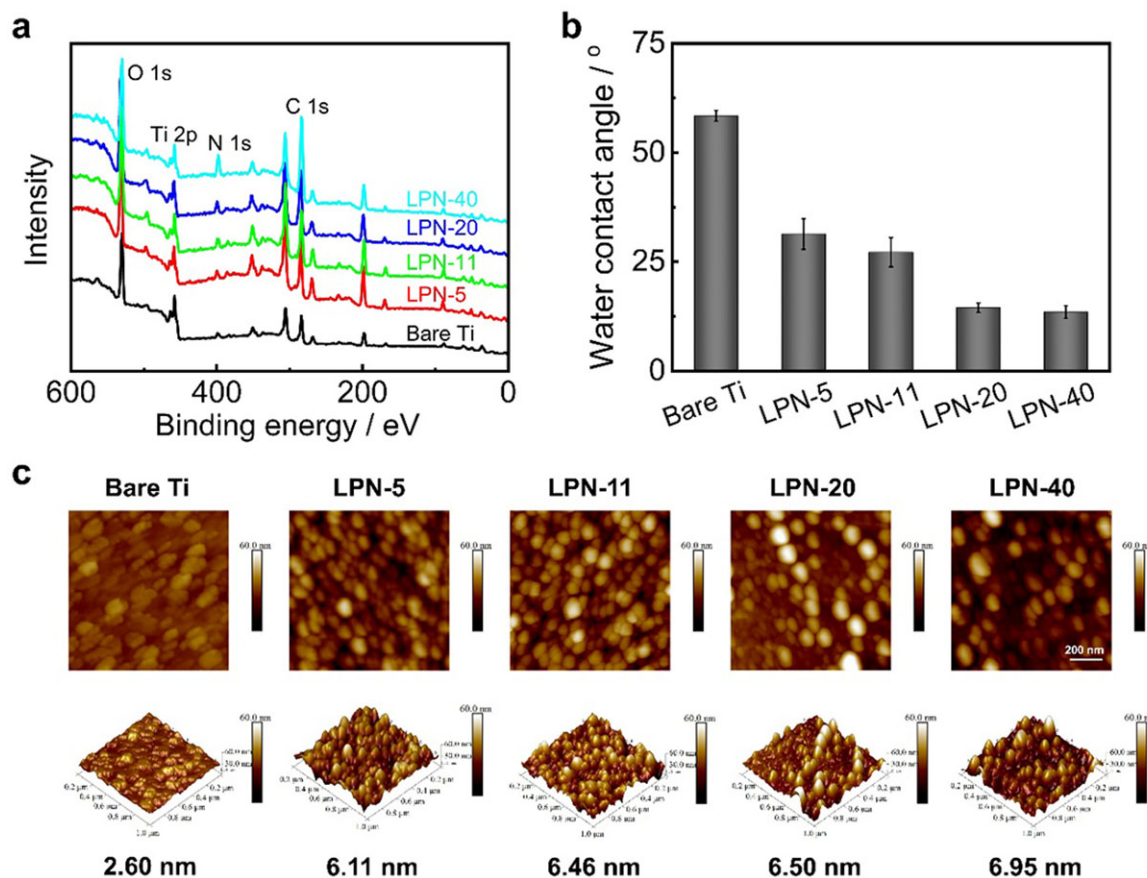


Figure 2. a) XPS spectra of the bare Ti and PNIPAM coatings. b) WCA of the bare Ti and PNIPAM coatings. c) 2D and 3D images of the surface morphology of the bare Ti and PNIPAM coatings. The scan size is $1 \mu\text{m} \times 1 \mu\text{m}$.

were observed without N peak (398 eV). In contrast, N peak attributed to PNIPAM was present in the case of LPN-x, indicating the adsorption of PNIPAM chains on the Ti surface.

Figure 2b shows WCA decreased with the molecular weight of PNIPAM. This is probably because more hydrophilic moieties are distributed on the top surface as the molecular weight increases. Namely, higher molecular weight is favorable for PNIPAM chains to adsorb on the Ti surface. However, LPN-20 (13°) and LPN-40 (12°) had WCA close to each other since the adsorption leveled off when PNIPAM molecular weight reached a critical value.

The morphology of the PNIPAM-coated Ti surface was observed by AFM after the QCM-D experiment (Figure 2c). The arithmetic mean roughness (R_a) of LPN-5, LPN-11, LPN-20, and LPN-40 were 6.11, 6.46, 6.50, and 6.95 nm, respectively, higher than that of the bare Ti surface (2.60 nm), indicating the adsorption of PNIPAM chains. The coating fabricated by LPN-20 or LPN-40 was thicker, so it was expected to serve for longer time. Considering that LPN-20 had good solubility in seawater, it was the best choice to prepare the coating.

The effects of PNIPAM concentration, temperature, flow rate, and seawater rinsing on the adsorption of LPN-20 were investigated by using QCM-D. Figure 3a shows $-\Delta f$ was less than 1000 Hz at a concentration below 0.5 mg mL^{-1} . As the concen-

tration increased, the adsorption increased, reflected in the large $-\Delta f$ and ΔD . When the concentration was above 1.0 mg mL^{-1} , the adsorption leveled off. Further increasing concentration did not change the adsorption.

Figure 3b shows that the adsorption increased with temperature in the range of 20–27 °C, indicated by the increase in $-\Delta f$ and ΔD . This is because the hydrophobicity of PNIPAM increases with temperature.^[30,51] However, LPN-20 precipitated at 30 °C since the temperature was above its LCST at 29 °C in artificial seawater (ASW) (Figure S3, Supporting Information),^[52] where PNIPAM chains formed non-uniform coating (Figure S4, Supporting Information). Note that a higher concentration of PNIPAM led to more adsorption on Ti, but precipitation happened when the concentration was above 0.2 mg mL^{-1} , so we chose 0.2 mg mL^{-1} for the experiments.

The flow rate greatly affected the adsorption. Figure 3c shows that $-\Delta f$ increased with the flow rate in the range we investigated, indicating that the flow is favorable to the self-fabrication of PNIPAM chains in a certain range. This is probably because the water flow disturbance at a high rate can promote the contacting of PNIPAM chains with the Ti surface.^[39] Certainly, the adsorbed PNIPAM chains would be washed out when the flow rate is above a critical value.

The standing of LPN-20 coating under a flow of seawater was investigated with QCM-D for 240 hours (10 days). Figure 3d

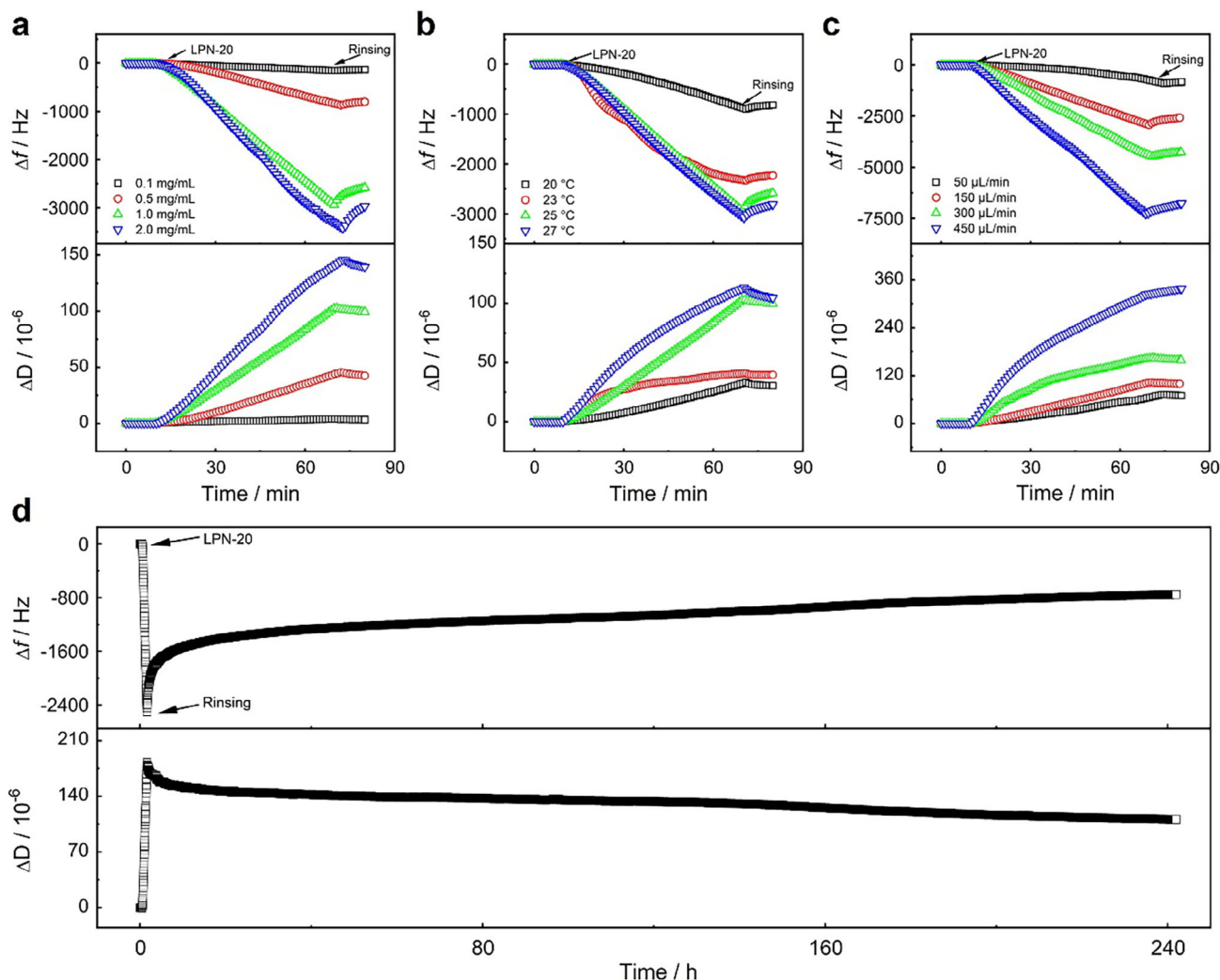


Figure 3. a) Time dependence of the frequency shift (Δf) or dissipation shift (ΔD) for the adsorption of LPN-20 at different concentrations on the Ti surface at 25 °C. b) Time dependence of Δf or ΔD for the adsorption of LPN-20 on the Ti surface as a function of temperature. c) Time dependence of Δf or ΔD for the adsorption of LPN-20 on the Ti surface at 25 °C as a function of flow rate. d) Time dependence of Δf or ΔD for adsorption of LPN-20 on the Ti surface at 25 °C for a long time.

shows that some PNIPAM chains adsorbed were washed away by ASW in the first 4 hours, reflected in the fact that $-\Delta f$ decreased from 2400 to 1600 Hz. However, even after 10 days of continuous flushing with ASW, $-\Delta f$ and ΔD remained ≈ 800 Hz and 1.4×10^{-4} , respectively, indicating that a certain amount of PNIPAM chains were still attached to the Ti surface, which would take the responsibility of antifouling. In short, PNIPAM can self-fabricate into a coating on the Ti surface. Unlike recently reported antifouling coatings such as degradable coating, hydrogel coating, and biomimetic superhydrophobic coating, which usually require a spray or spin coating process, the current method provides a facile solution of protecting seawater pipeline with limited space for coating application.^[26,53] Actually, polymers containing phosphoric acid and lysine groups are able to adsorb onto the Ti surface to form a coating due to their coordination interactions with Ti.^[54]

2.2. Antifouling Properties

It is recognized that the formation of a conditioning film composed of proteins, glycoproteins, and other substances on the surface is the first step for marine biofouling.^[55,56] Therefore, protein resistance makes a significance for marine antifouling. Protein adsorption on the LPN-20 coating was evaluated by QCM-D measurements (Figure 4a,b). For fibrinogen, bovine serum albumin (BSA), and lysozyme on bare Ti surface, $-\Delta f$ values were 324, 87, and 77 Hz, respectively. This is due to Ti's excellent biocompatibility. In contrast, almost no adsorption was on the LPN-20-coated Ti surface, reflected in the zero shifts of Δf . Accordingly, the PNIPAM coating had good nonspecific protein resistance.

Figure 4c shows the fluorescence photographs of the suspension of gram-negative bacteria (*E. coli*), gram-positive bacteria (*S. aureus*), and marine bacteria (*P. sp.*) incubated on LPN-20-coated

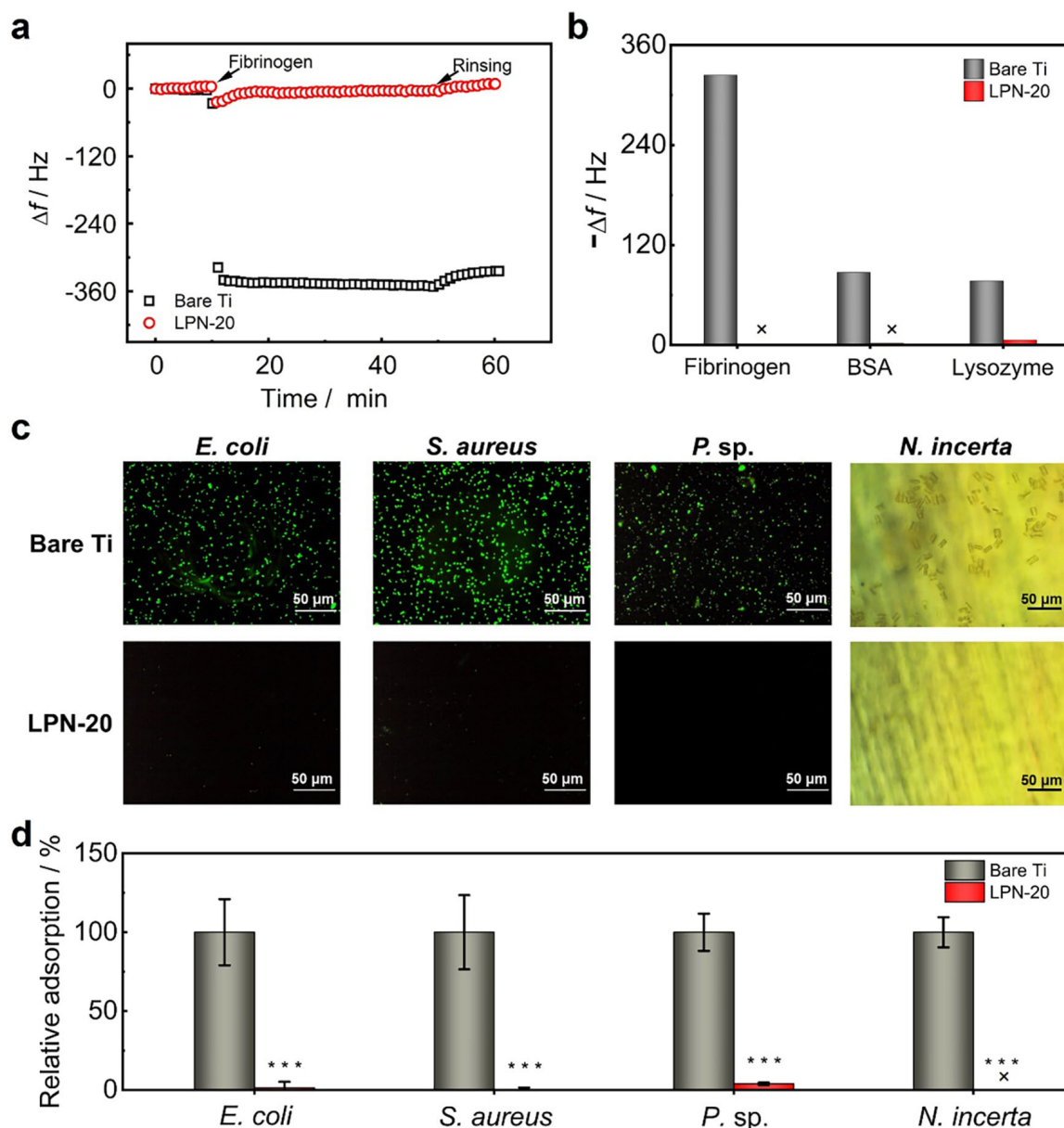


Figure 4. a) Time dependence of the frequency shift (Δf) for the fibrinogen protein adsorbed on the surface at 25 °C. b) Frequency shift (Δf) of fibrinogen, BSA, and lysozyme adsorbed on the surface before and after coating of LPN-20. c) Fluorescence photographs of the suspension of gram-negative bacteria (*E. coli*), gram-positive bacteria (*S. aureus*), marine bacteria (*P. sp.*), and diatom (*N. incerta*) incubated on the Ti surface before and after coating of LPN-20. d) Relative adsorption of *E. coli*, *S. aureus*, *P. sp.*, and *N. incerta*, respectively.

Ti surface. It is not surprising to observe numerous bacteria on the bare Ti surface as it has excellent biocompatibility and is susceptible to bacterial adhesion. However, for each type of microorganism, the number sharply decreased on the LPN-20-coated Ti surface, indicating the good antibacterial properties of the coating. Figure 4c shows the microscopic images of the suspension of diatom (*N. incerta*) incubated on the LPN-20-coated Ti surface. Only a few diatoms were observed on the LPN-20-coated Ti surface, though there were many on the bare Ti surface. Namely, PNIPAM coating was able to inhibit the adhesion of the diatom. Through Image-J analysis, the relative adsorption ratio of bacteria

and diatom on LPN-20-coated Ti was $\approx 1\%$ (Figure 4d). The good biofouling resistance of the coating is attributed to the hydrated layers of PNIPAM chains.

It should be noted that the hydration of PNIPAM is quite different at a temperature below or above its LCST (Figure S4, Supporting Information). Similar phenomena of temperature sensitive of PNIPAM based systems have been reported, such as PNIPAM-capped Au nanoparticles and cross-linked polymer coupled with PNIPAM.^[32,33] We examined the anti-biofouling properties of PNIPAM coating at a temperature above the LCST (Figure S5, Supporting Information). Considering that the

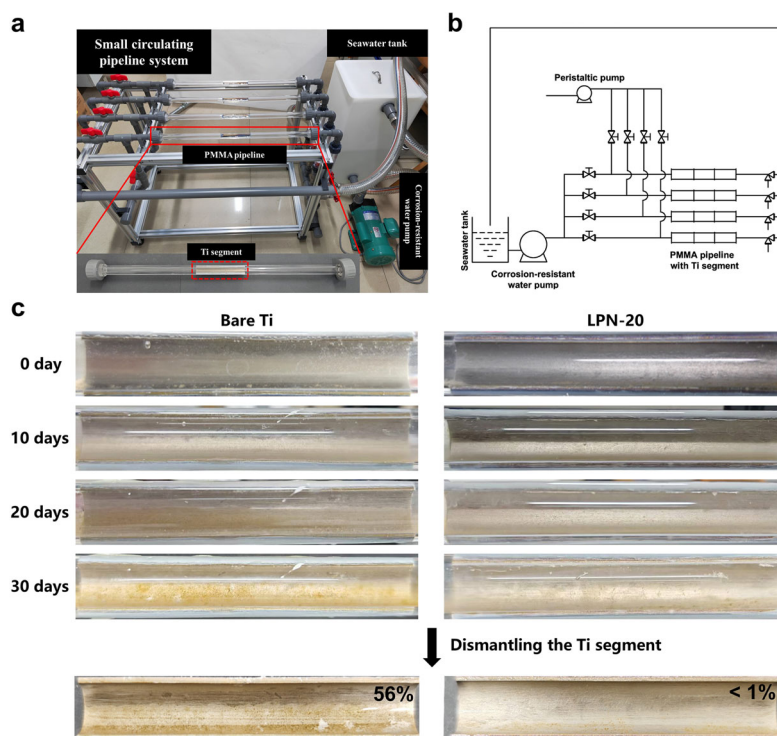


Figure 5. a) Image of the small circulating pipeline system. b) Schematic diagram of the small circulating pipeline system. c) Photos of the biofilm growth on the Ti segment surface in 0, 10, 20, and 30 days, and photos of the biofilm growth on the surface of the Ti segment after dismantling.

temperature in seawater pipelines is generally below 30 °C, the experiment was conducted at 30 °C. The number of bacteria or diatoms on the LPN-20-coated Ti surface significantly decreased in comparison with that on the bare Ti surface. This is because PNIPAM chains are not completely dehydrated and trapped free water molecules at a temperature above LCST,^[57] the bonded and free water molecules can form a dynamic surface which resists the biofouling.^[27,56]

The long-term antibacterial performance of LPN-20 coating was examined. Figure S6 (Supporting Information) shows the fluorescence photographs of the suspension of marine bacteria (*P. sp.*) incubated on the bare Ti surface and LPN-20-coated Ti surface for 1, 3, 7, and 10 days. The amount of bacteria on the bare Ti surface remarkably increased with time, and it was completely covered by bacteria in 10 days. In contrast, only a few bacteria were on LPN-20-coated Ti surface even after 10 days, indicating that the coating has relatively long-term antifouling efficacy.

The regeneration efficiency of the LPN-20 coating was also investigated. It was regenerated after immersion for 10 days, and its antifouling performance was evaluated via bacterial assay (Figure S7, Supporting Information). Almost no bacteria were observed on LPN-20-coated Ti surface, whereas a large number of bacteria were adhered on the bare Ti surface, indicating that the coating has good regeneration efficiency.

Owing to the osmotic pressure and water flow, the coated PNIPAM chains would be washed away upon the flow of seawater. We examined the antifouling performance of LPN-20-coated Ti pipeline by using a small circulating pipeline system (Figure 5a), where the seawater flow rate was set to 0.3 m s⁻¹, and LPN-20 solution with concentration of 1.0 g L⁻¹ was added once every

10 days. A large-scale bacterial biofilm formed on the inner wall of Ti pipe without the coating in 30 days, but almost no bacterial biofilm was observed in the LPN-20 coated pipeline, demonstrating its good antifouling performance (Figure 5c). For the Ti segment, biofilm covered over 56% of the area of the bare Ti surface, but it did less than 1% of the area of the LPN-20-coated Ti surface.

2.3. Effect of Coating on Anti-Corrosion of Ti Surface

Ti is advantageous for its high corrosion resistance. Whether PNIPAM coating influences its corrosion resistance is our concern. Figure 6a shows that the low-frequency impedance modulus ($|Z|_{0.01 \text{ Hz}}$) for the control sample after immersion in ASW decreased in the Bode plot. Meanwhile, the capacitance semicircle diameter in the Nyquist plot decreased, indicating a certain degree of corrosion resistance loss (Figure 6b). In contrast, the decrease in either $|Z|$ value or the diameter regarding the LPN-20-coated Ti surface was much lower. As shown in Figure 6c,d, the LPN-20 sample exhibited a lower corrosion current density, a higher corrosion potential, and a lower corrosion rate in comparison with the control. These facts indicate that PNIPAM coating did not reduce but slightly improved the corrosion resistance of Ti surface since it acted as the protective layer.

3. Conclusion

We have developed a self-fabricated coating based on poly(*N*-isopropylacrylamide) (PNIPAM) to combat marine biofouling

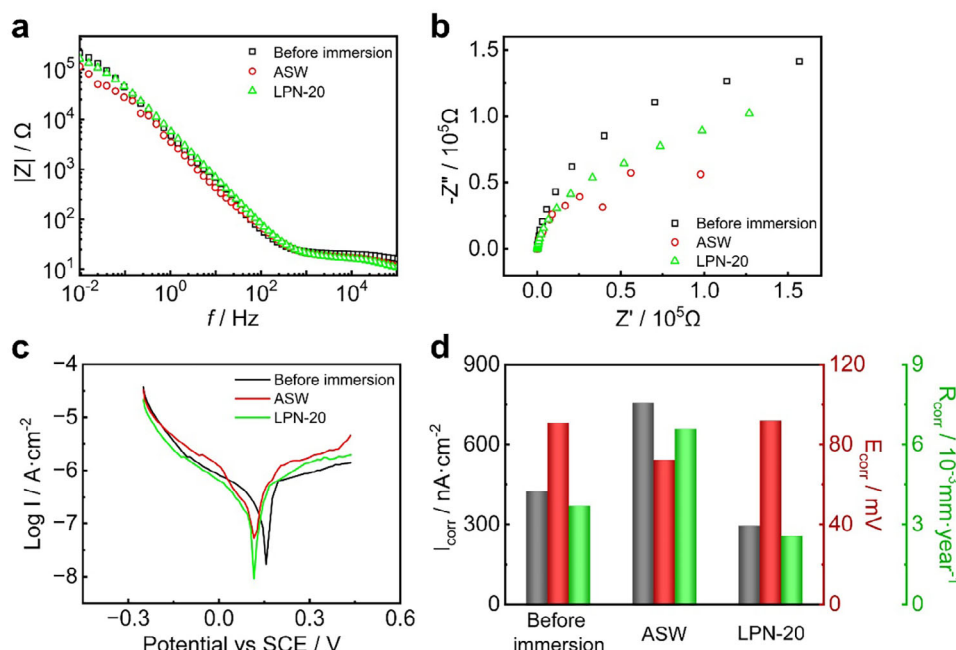


Figure 6. a) Bode and b) Nyquist plots obtained from electrochemical impedance spectroscopy (EIS) measurements for Ti after immersion for 7 days. c) Tafel curves of Ti after immersion for 7 days. d) Tafel curve fitting parameters.

in titanium seawater pipeline systems. PNIPAM chains self-fabricate into a coating on the inner wall of the pipe due to the coordination interactions of the nitrogen atoms with Ti and the hydrophobic interactions. Such a self-fabricated coating exhibits good anti-biofouling, while it does not change the corrosion resistance of titanium. The coating can last for more than 10 days at a flow velocity of 0.3 m s^{-1} and continue to function after regeneration with a PNIPAM solution. This study provides a facile and feasible method to protect titanium pipelines from marine biofouling.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

corrosion, marine biofouling, poly(*N*-isopropylacrylamide), quartz crystal microbalance, titanium

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